DESIGN AND INNOVATION

INTRODUCTION

In a world driven by creativity and rapid technological evolution, Unmanned Aerial Vehicles (UAVs) have become platforms for both engineering precision and imaginative design. The Design & Innovation Challenge celebrates this fusion by encouraging participants to build drones beyond conventional norms, exploring unique structures, materials, and functionalities. It empowers innovators to transform bold ideas into tangible aerial systems that redefine performance, aesthetics, and technological possibilities.

Mohammed Ali
Outreach Head
ali@dronotics.in

Vaibhav Katariya
Head Events
vaibhav@dronotics.in

Yasharth Singh
Management Head
yasharth@dronotics.in

Rakshit Suneja
Creative Head
rakshit@dronotics.in

PROBLEM STATEMENT

Design and develop an innovative UAV-based drone system that solves a real-world problem through a blend of creativity and engineering. Participants must present more than just an idea by showcasing a working prototype, simulation, or scaled model. The solution should clearly demonstrate functionality, feasibility, and how the drone effectively performs its intended task in practical scenarios.

The drone must integrate unique design elements with technical depth, such as advanced structures, payload mechanisms, or sensor-based applications. Teams should support their work with proper documentation, research insights, or analysis. The final outcome should highlight innovation, efficiency, and scalability, ensuring the proposed UAV system has meaningful real-world relevance and potential impact.

ABSTRACT SUBMISSION ROUND

The participating teams must submit an abstract of their drone's concept, in proper documented form. This should include its configuration, design, features, estimated cost, and an approach to tackle the problem statement. The teams must compile all this information in a PDF file. If any extra files are to be included, the submission must be in the form of a ZIP file. Please note that the files submitted should NOT be password-protected or corrupted. In either of the cases, the submissions would be directly rejected.

GUIDELINES FOR ABSTRACT SUBMISSION

Rough sketches of your idea including the Design Report of your model. The abstract must specify the use of software and the mechanism used in the design analysis of the drone.

- USPs (Unique Selling Propositions) of your design and innovations made by you should be mentioned in the report and mention how they are going to tackle the specific challenges.
- An abstract of the idea needs to be uploaded on the Dronotics website before the deadline.
- The abstract should not exceed 500 words and should include the following details of all team members along with the Team.

Cover Page Details:

The cover page of the submission must include the following information:

- Team Name
- Team Leader Details:
  - Name of Team Leader
  - College / Institute Name
  - Year of Study
  - Email ID
  - Phone Number
- Team Members:
  - Names of all team members

File Naming Convention:

The file must be submitted in the following format:

DNI_<TeamName>_<LeaderName>_abstract.pdf

Example: DNI_ABC_AmanGandotra_abstract.pdf

Submission Email

The abstract/design report must be mailed to:
dronoticsjiit128@gmail.com

The Subject of email should be in this format:
Design and Innovation - <TeamName> - <LeaderName> Abstract
Example: Design and Innovation - ABC - Aman Gandotra Abstract

Important Note

- Only PDF format submissions will be accepted.
- Files must not be corrupted or password-protected.
- Any submission not following the naming convention may be rejected.

Note: The organizers reserve all rights to change any or all of the above rules.